ASTRA environment-split clustering: cosmological forecasts with emulators and comparison with density split

Hugo Nguyen
DESI

Why environments?

The two-point correlation function (2PCF) captures all the information of a Gaussian field, but the late-time Universe is not Gaussian: matter collapses into halos, filaments and nodes, and empties voids. **Environment statistics** recover part of the missing information by measuring clustering separately in dense and empty regions.

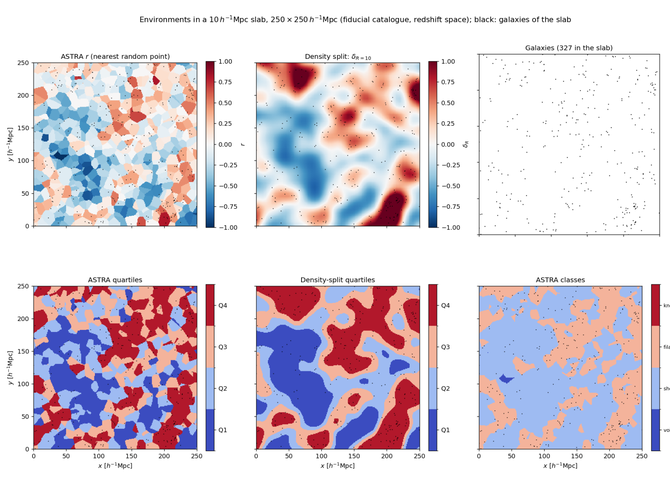
Goal: quantify how much **ASTRA** environment statistics improve constraints on ω_b, $\omega_{\rm cdm}$, n_s, σ_8, w_0, w_a over the 2PCF, and compare with **density split**, the reference method.

Two ways to define an environment

ASTRA: Delaunay tessellation of galaxies + random points; each point gets

$$r = \frac{N_{\rm gal} - N_{\rm rand}}{N_{\rm gal} + N_{\rm rand}}$$

from its neighbours. No smoothing scale: adaptive, local.
Density split: galaxy density smoothed on $10\,h^{-1}$Mpc. Points are ranked into quartiles Q1 (voids) to Q4 (clusters), or into ASTRA classes (void, sheet, filament, knot).



Environments in a 250 × 250 × 10 h^{-1}Mpc slab; black dots: galaxies.

Data and method

AbacusSummit	DESI-like LRG mocks, 85 cosmologies: measurements, emulator
DESI official mocks	1786 independent boxes: covariance
Quijote	500 paired simulations per parameter: clean derivatives

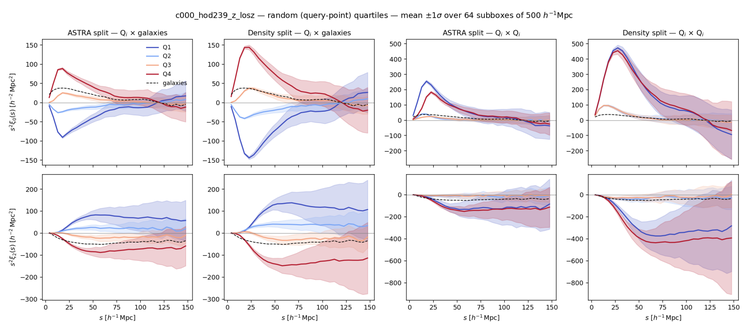
Auto- and cross-correlations of the quartiles (monopole + quadrupole) → covariance + derivatives → **Fisher forecasts**. A **neural-network emulator** of all the statistics gives derivatives at fixed galaxy–halo connection, and an **MCMC** tests the full inference.

Emulator accuracy



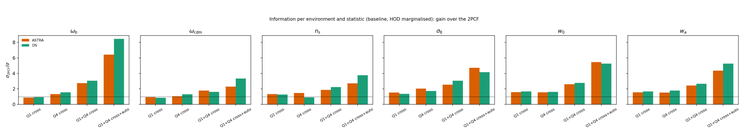
Error on 8 cosmologies never seen in training: $\sim 0.5\sigma$ of a $2\,h^{-1}$Gpc box, ten times below the error of the volume used for the forecasts (dashed).

Clustering by environment



Monopole (top) and quadrupole (bottom) of the random-quartile cross- and auto-correlations. ASTRA signals are $\sim 1.6\times$ weaker, but information is about signal *and* noise.

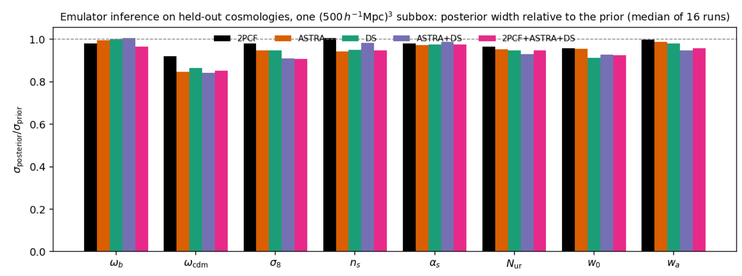
Gain over the 2PCF



$\sigma_{\rm 2PCF}/\sigma$ per environment and statistic. Each quartile alone is about as informative as the 2PCF; combined, and with auto-correlations, they are several times better.

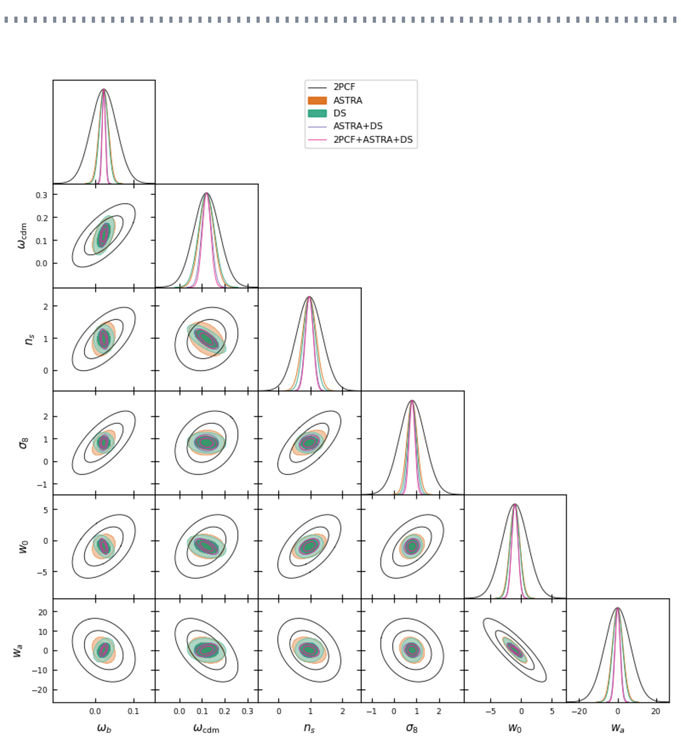
	vs 2PCF	DS / ASTRA
gain on $\omega_b, \ldots, w_a$	×1.6–3.1	0.8–1.1
ASTRA + DS combined	further	×1.2–2.4

Inference test (MCMC)



Posterior width / prior width on held-out cosmologies. The posteriors are unbiased and calibrated; one $(500\,h^{-1}\text{Mpc})^3$ volume is too small to discriminate the statistics, a full box is limited by the emulator error.

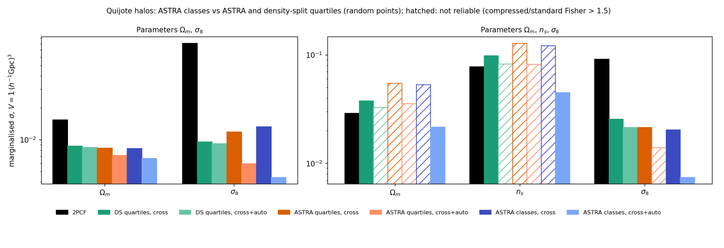
Fisher contours



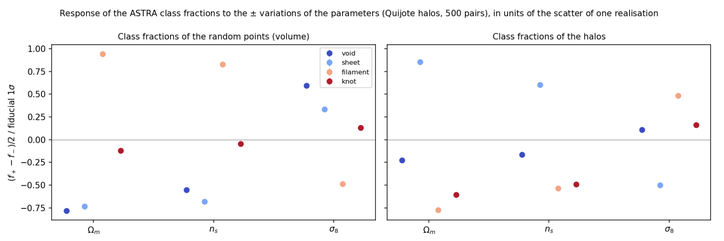
2PCF, ASTRA, density split and combinations, one $(500\,h^{-1}\text{Mpc})^3$ volume, galaxy–halo connection marginalised.

ASTRA cosmic-web classes

With fixed thresholds, voids and knots hold $< 1\%$ of the points: two dominant classes (sheets, filaments). Yet on Quijote, their auto- and cross-correlations are the **most constraining statistic**:



σ_8 is **2–3 times** better constrained than with density split, robustly across data compressions.



With fixed thresholds, the volume of each class responds to cosmology, which equal-size quartiles remove by construction.

What we learnt about the method

- Emulator suites draw a different galaxy–halo connection in each cosmology: finite differences measure it, not cosmology → emulator.
- A covariance from one simulation box biased the comparison → 1786 independent boxes.
- Derivatives of environment statistics of sparse tracers are noisy → two bracketing Fisher estimators identify reliable forecasts.
- ASTRA adapts its scale to the density: model it at the density of the data.

Conclusions

- ASTRA environment statistics constrain cosmology $\sim 2\times$ **better than the 2PCF**.
- As informative as density split, and **complementary**.
- ASTRA **classes** constrain σ_8 2–3× better than density split (Quijote).
- Emulator + MCMC pipeline validated, ready for larger training sets.

Code: `github.com/Hugoooat/DESI_astra` (branch `astra_hugo_v2`)

LaTeX TikZposter